



University of Stuttgart
Collaborative
Artificial Intelligence



e l l i s
European Laboratory for Learning and Intelligent Systems



University of Stuttgart
Germany

Machine Perception and Learning for Collaborative Intelligent Systems

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WS 2025/2026

Collaborative Artificial Intelligence Group, University of Stuttgart

www.collaborative-ai.org 

Transformers in Computer Vision

- Vision Transformer
- End-to-end Object Detection in Visual Transformers
- Self-supervised Vision Transformers

Multimodal Learning

Motivation of Multimodal Learning

Multimodal Representation Learning

Multimodal Alignment

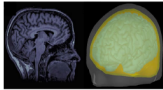
Multimodal Reasoning

Multimodal Learning

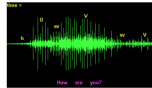
Motivation of Multimodal Learning



Psychology



Medical



Speech



Vision



Language



Multimedia

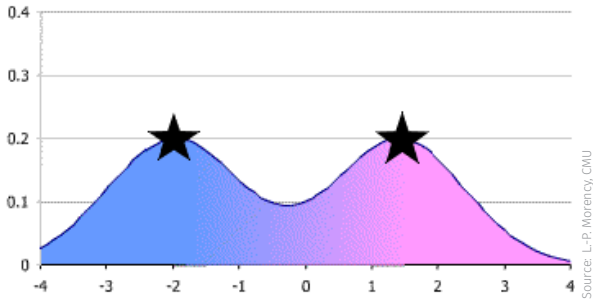


Robotics

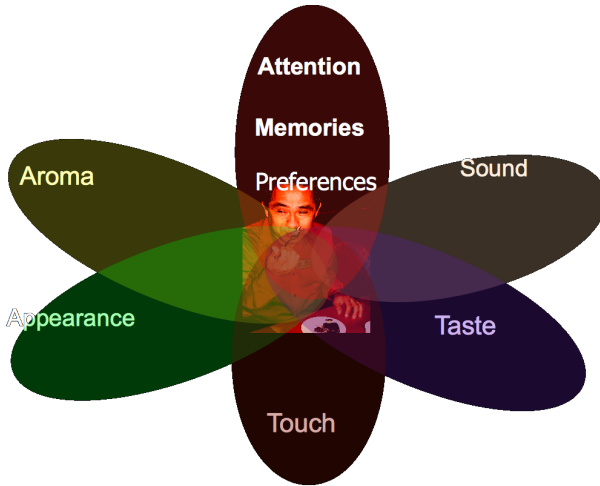
A photograph of a blackboard filled with complex mathematical equations, representing learning.

Learning

Source: L-P. Morency, CMU



- Multiple modes, i.e., distinct “peaks” (local maxima) in the probability density function



Source: L.-P. Morency, CMU

Language

- **Lexicon**
 - Words
- **Syntax**
 - Part-of-speech
 - Dependencies
- **Pragmatics**
 - Discourse acts

Acoustic

- **Prosody**
 - Intonation
 - Voice quality
- **Vocal expressions**
 - Laughter, moans

Visual

- **Gestures**
 - Head gestures
 - Eye gestures
 - Arm gestures
- **Body language**
 - Body posture
 - Proxemics
- **Eye contact**
 - Head gaze
 - Eye gaze
- **Facial expressions**
 - FACS action units
 - Smile, frowning

Touch

- **Haptics**
- **Motion**

Physiological

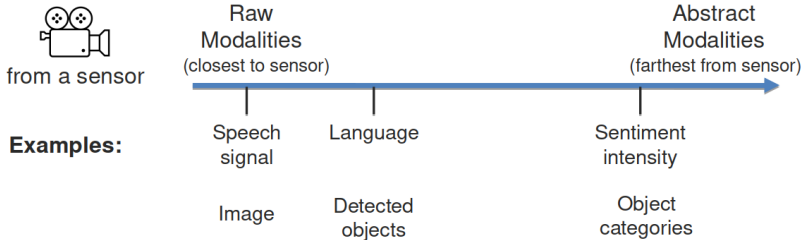
- **Skin conductance**
- **Electrocardiogram**

Mobile

- **GPS location**
- **Accelerometer**
- **Light sensors**

Source: L.-P. Morency, CMU

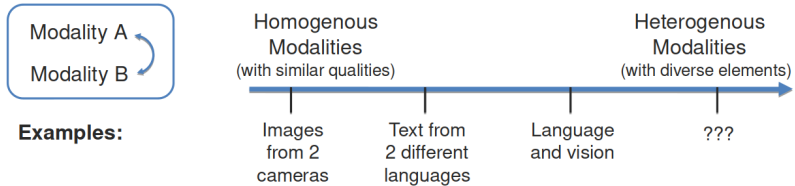
Modality refers to the way in which something is expressed or perceived



Source: L.-P. Morency, CMU

Multimodal: with multiple modalities

Heterogeneity: Information present in different modalities will often show diverse qualities and elements



Source: L.-P. Morency, CMU

Multimodal Machine Learning is the study of computer algorithms that learn and improve through the use and experience of multimodal data

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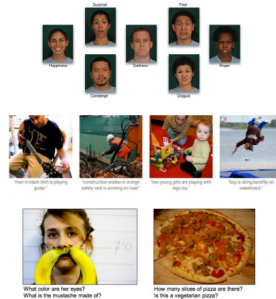
Multimodal Artificial Intelligence studies computer agents that are able to demonstrate intelligence capabilities – such as understanding, reasoning, and planning – through multimodal experiences and data

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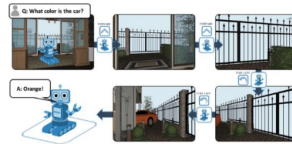
Multimodal is the scientific study of heterogeneous and interconnected (connected + interacting) data

- Affect recognition
 - Emotion
 - Sentiment
 - Personality traits
- Media description
 - Image & video captioning
- Multimodal question answering
 - Image & video QA
 - Visual reasoning
 - Visual dialog

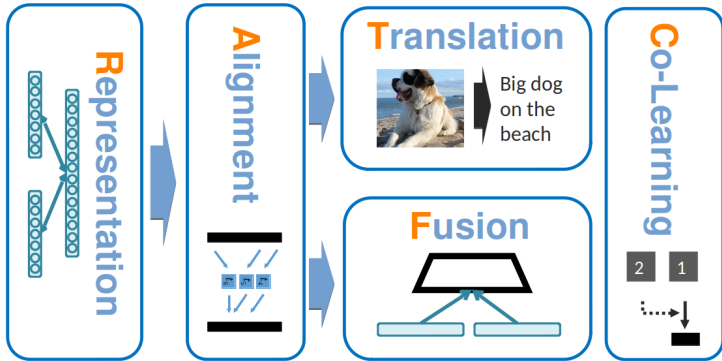


Source: Unknown

- Multimodal navigation
 - Language-guided navigation
 - Autonomous driving
- Event recognition
 - Action recognition
 - Segmentation
- Multimedia information retrieval
 - Content-based/Cross-media



Source: Unknown



Source: L.-P. Morency, CMU

Slide source: [Baltrušaitis et al., 2018]

Representation

- Joint
 - *Neural networks*
 - *Graphical models*
 - *Sequential*
- Coordinated
 - *Similarity*
 - *Structured*

Translation

- Example-based
 - *Retrieval*
 - *Combination*
- Model-based
 - *Grammar-based*
 - *Encoder-decoder*
 - *Online prediction*

Alignment

- Explicit
 - *Unsupervised*
 - *Supervised*
- Implicit
 - *Graphical models*
 - *Neural networks*

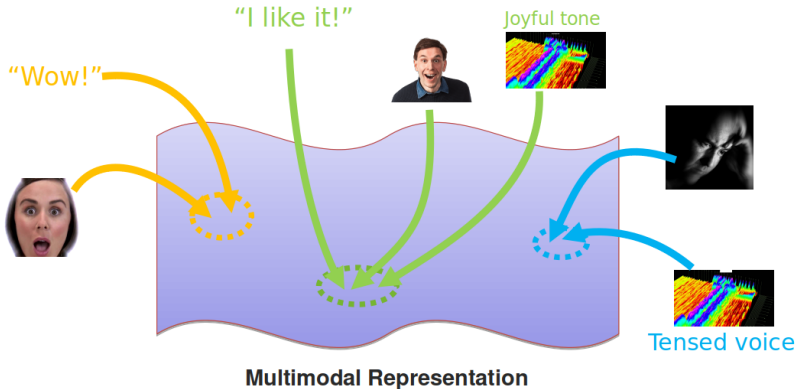
Fusion

- Model agnostic
 - *Early fusion*
 - *Late fusion*
 - *Hybrid fusion*
- Model-based
 - *Kernel-based*
 - *Graphical models*
 - *Neural networks*

Co-learning

- Parallel data
 - *Co-training*
 - *Transfer learning*
- Non-parallel data
 - *Zero-shot learning*
 - *Concept grounding*
 - *Transfer learning*
- Hybrid data
 - *Bridging*

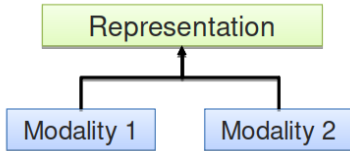
Source: L.-P. Morency, CMU



Source: L.-P. Morency, CMU

Definition: Learning representations that reflect cross-modal interactions between individual elements across different modalities.

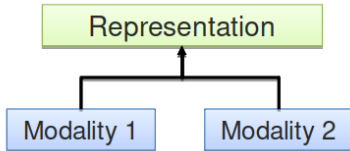
Joint representations



Source: L.-P. Morency, CMU

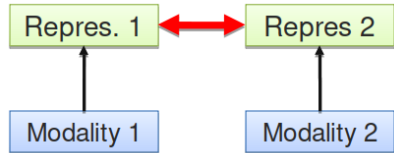
Definition: Learning representations that reflect cross-modal interactions between individual elements across different modalities.

Joint representations



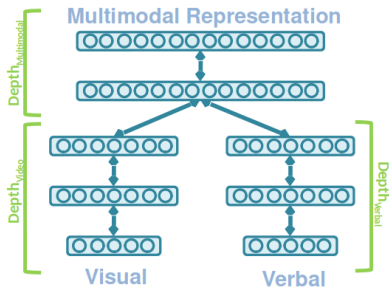
Source: L.-P. Morency, CMU

Coordinated representations



Source: L.-P. Morency, CMU

- Audio-visual speech recognition
 - Bimodal Deep Belief Network [Ngiam et al., 2011]
- Image captioning
 - Multimodal Deep Boltzmann Machine [Srivastava et al., 2012]



Source: L-P. Morency, CMU

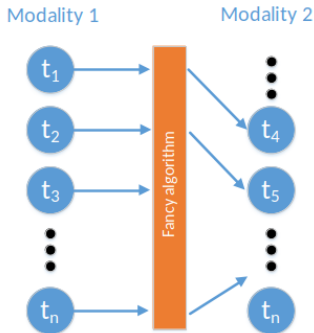
- Audio-visual emotion recognition
 - Deep Boltzmann Machine [Kim et al., 2013]

Multimodal Vector Space Arithmetic



Source: [Kiros et al., 2014]

Definition: Identify the direct relations between (sub)elements from two or more different modalities



Source: L.-P. Morency, CMU

Explicit alignment

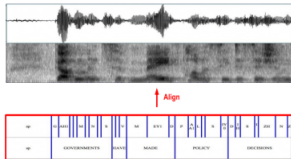
- The goal is to directly find correspondences between elements of different modalities

Implicit alignment

- Uses internally latent alignment of modalities in order to better solve a problem

Finds correspondences between modalities, i.e. **the alignment is part of the loss function**. Examples:

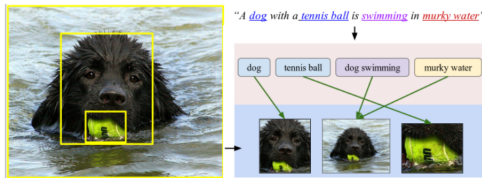
- Aligning speech signal to a transcript
- Aligning two out-of sync sequences
- Language grounding



Source: Unknown

Uses an internal (latent) alignment of modalities. Examples:

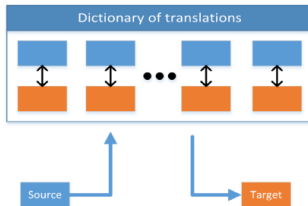
- Machine Translation
- Cross-modal retrieval
- Image & Video Captioning
- Visual & Video Question Answering
- Visual Dialog



Source: [Karpathy et al., 2014]

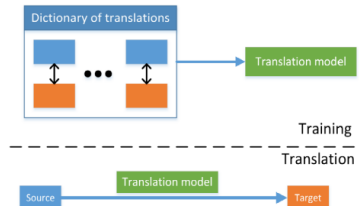
Definition: Change (“translate”) data from one modality to another, where the translation relationship can often be open-ended or subjective

Example-based



Source: L.-P. Morency, CMU

Model-driven



Source: L.-P. Morency, CMU

a person jogs a few steps



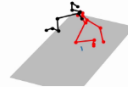
**a person jogs a
few steps**

A person steps forward then turns
around and steps forwards again.



**A person steps forward then
turns around and steps
forwards again.**

A kneeling person raises their arms to
the sides and stands up.

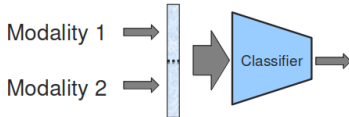


**A kneeling person raises
their arms to the sides and
stand up.**

Source: [Ahuja and Morency, 2019]

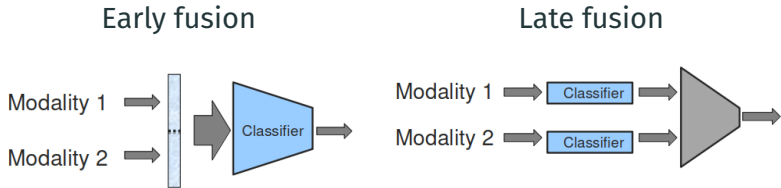
Definition: Join information from two or more modalities to perform a prediction task

Early fusion



Source: L.-P. Morency, CMU

Definition: Join information from two or more modalities to perform a prediction task



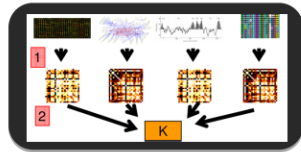
Source: L.-P. Morency, CMU

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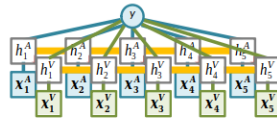
Definition: Join information from two or more modalities to perform a prediction task

Model-based (intermediate) approaches

- Deep neural networks
- Kernel-based methods
- Graphical models

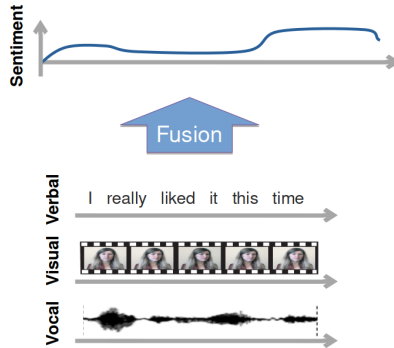


Multiple kernel learning



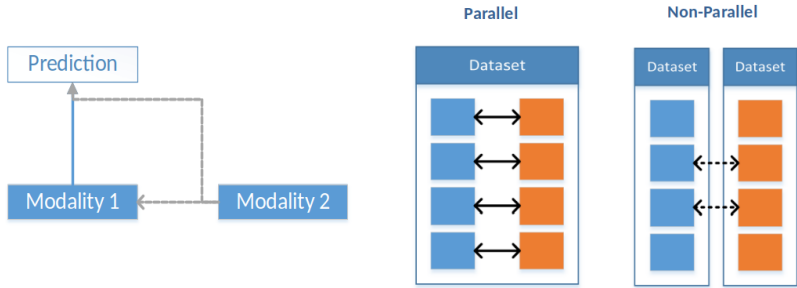
Multi-View Hidden CRF

Source: L.-P. Morency, CMU

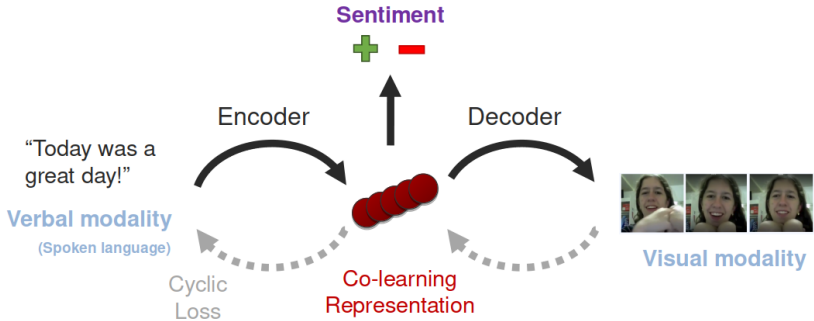


Source: L.-P. Morency, CMU

Definition: Transfer knowledge between modalities, including their representations and predictive models

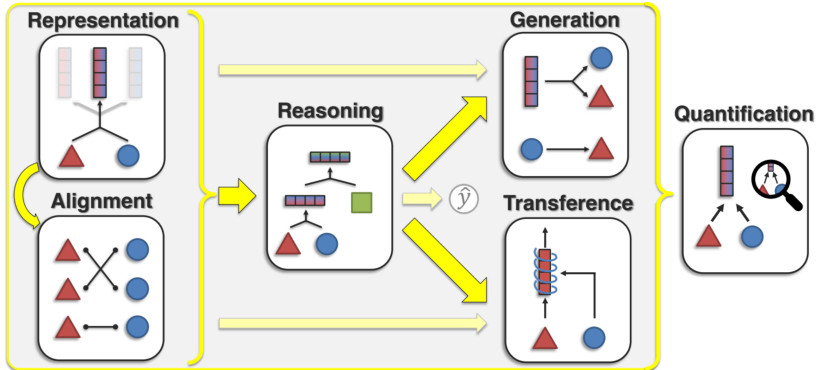


Source: L.-P. Morency, CMU



Source: L.-P. Morency, CMU

Slide source: [Pham et al., 2019]



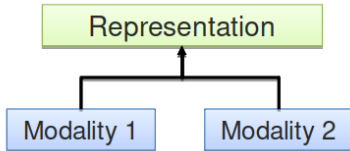
Learn more at CVPR 22' Tutorial [↗](#)

Multimodal Learning

Multimodal Representation Learning

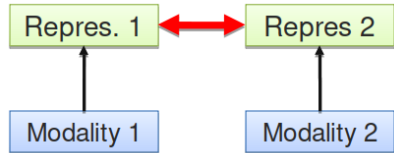
Definition: Learning representations that reflect cross-modal interactions between individual elements across different modalities.

Joint repres. (fusion)



Source: L.-P. Morency, CMU

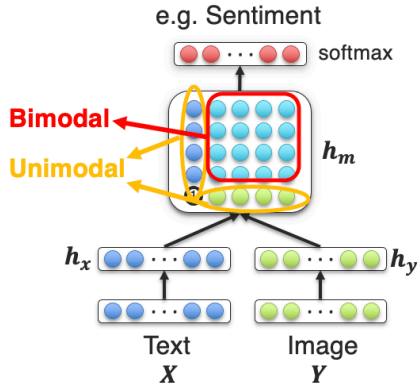
Coordinated repres.



Source: L.-P. Morency, CMU

Models both unimodal and bimodal interactions:

$$h_m = \begin{bmatrix} h_x \\ 1 \end{bmatrix} \otimes \begin{bmatrix} h_y \\ 1 \end{bmatrix} = \begin{bmatrix} h_x & h_x \otimes h_y \\ 1 & h_y \end{bmatrix}$$



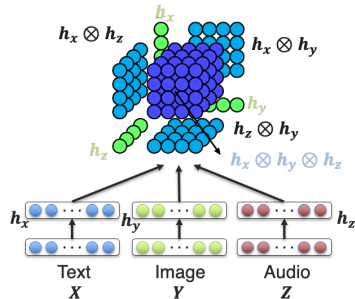
Source: L-P. Morency, CMU

Slide source: [Zadeh et al., 2017]

- Can be extended to three modalities:

$$h_m = \begin{bmatrix} h_x \\ 1 \end{bmatrix} \otimes \begin{bmatrix} h_y \\ 1 \end{bmatrix} \otimes \begin{bmatrix} h_z \\ 1 \end{bmatrix}$$

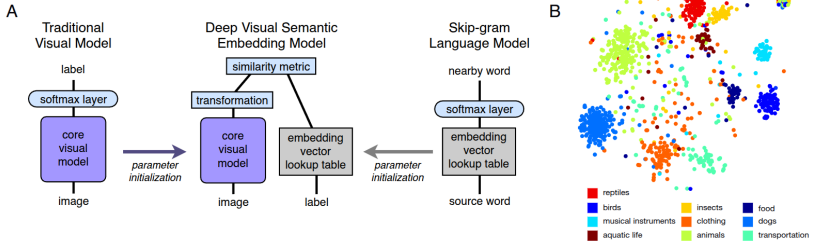
- Explicitly models **unimodal**, **bimodal**, and **trimodal** interactions



Source: L-P. Morency, CMU

Slide source: [Zadeh et al., 2017]

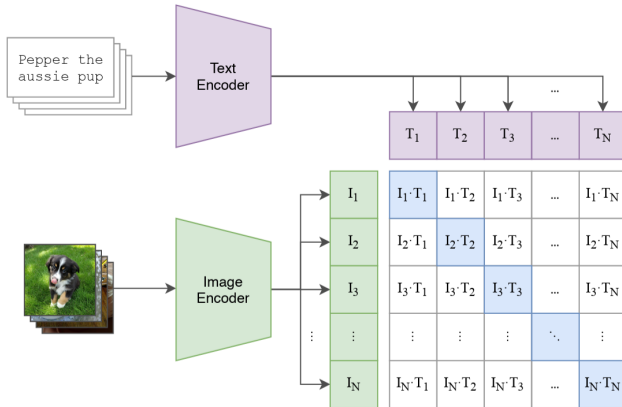
- DeViSE: A Deep Visual-Semantic Embedding Model[Frome et al., 2013]
- Earliest work on multimodal representation learning
- Lightweight in **word embedding** and **modality interaction**, but heavy in the **vision encoder**



Source: [Frome et al., 2013]

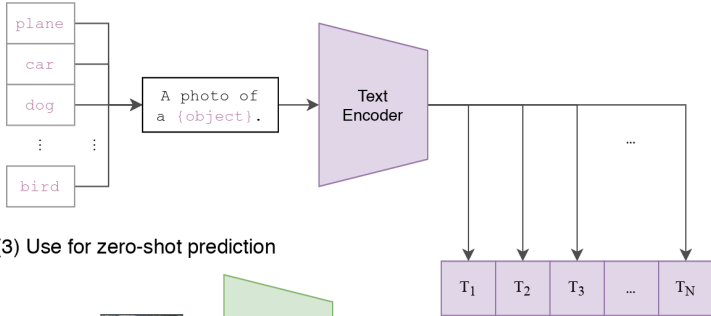
Diagonal values positive, other negative

(1) Contrastive pre-training

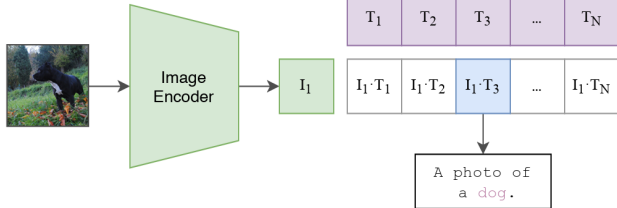


Source: [Radford et al., 2021]

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



Source: [Radford et al., 2021]

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t            - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]
```

Source: [Radford et al., 2021]

```
# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss    = (loss_i + loss_t)/2
```

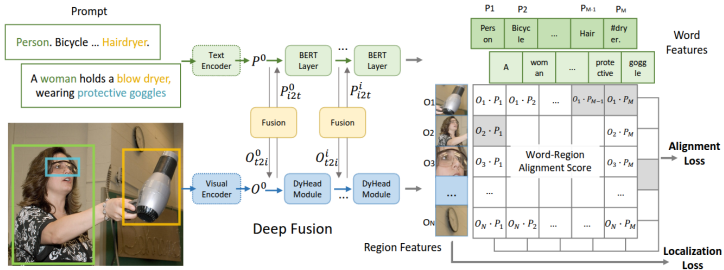
Source: [Radford et al., 2021]

- **Joint** representation learning with cosine similarity
- Lightweight in **modality interaction**, but heavy in both **language and vision encoder**
- Got rid of categorical labels, able for **zero-shot prediction**
- Prompt engineering is important
- Learnt robust features that generalised well

	Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet		76.2	76.2	0%
ImageNetV2		64.3	70.1	+5.8%
ImageNet-R		37.7	88.9	+51.2%
ObjectNet		32.6	72.3	+39.7%
ImageNet Sketch		25.2	60.2	+35.0%
ImageNet-A		2.7	77.1	+74.4%

Source: [Radford et al., 2021]

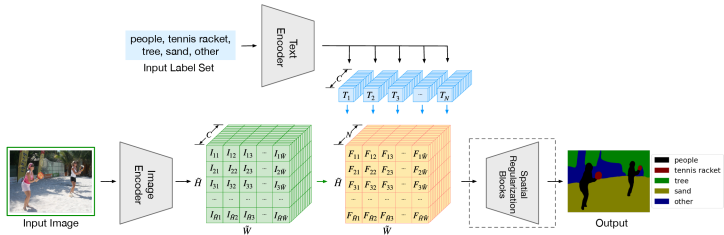
Grounded Language-Image Pre-training [Li et al., 2022c]



Source: [Li et al., 2022c]

- Using large training datasets in vision grounding
- Allowing **zero-shot** object detection, i.e. open vocabulary

Language-Driven Semantic Segmentation [Li et al., 2022a]



Source: [Li et al., 2022a]

- Extending the CLIP model to semantic segmentation
- Freezing text encoder weights from CLIP, supervised learning

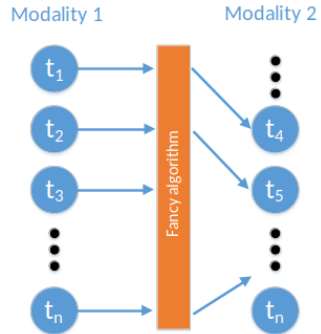
Multimodal Learning

Multimodal Alignment

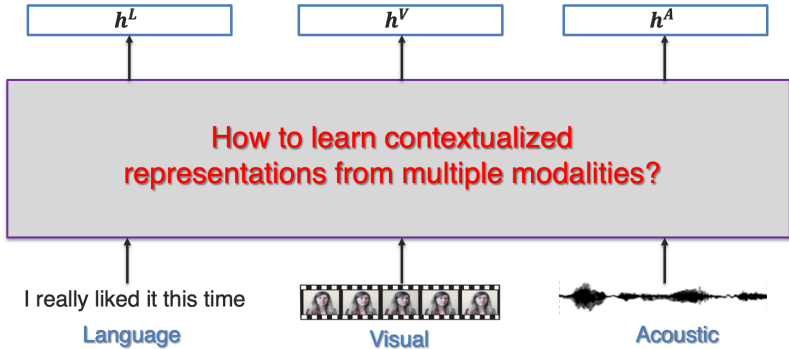
Goal: Find relationships/correspondences between two or more modalities

Two types:

- **Explicit** – alignment is the task in itself
- **Implicit / Latent** – alignment helps when solving a different task (e.g. using “Attention” module)



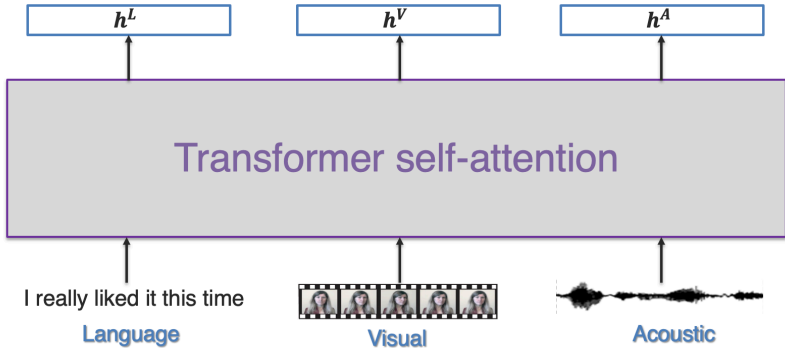
Source: L-P. Morency, CMU



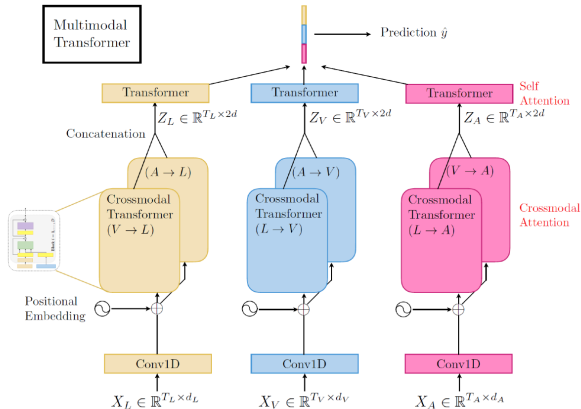
Source: L.-P. Morency, CMU

Look at pairwise interactions between modalities

Contextualised Multimodal Embeddings

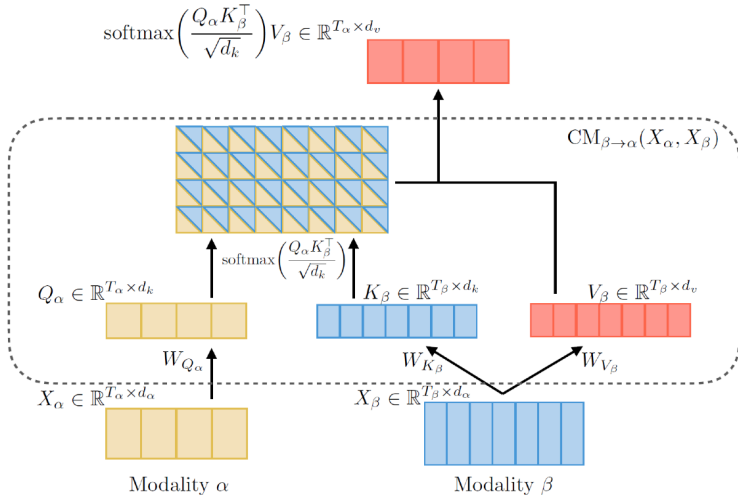


Source: L.-P. Morency, CMU



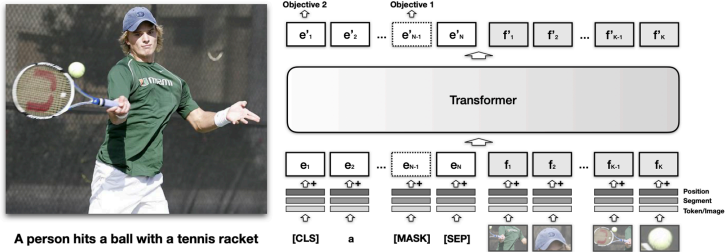
Source: [Tsai et al., 2019]

How should we connect query, key, and value in a cross-modal transformer?



Source: [Tsai et al., 2019]

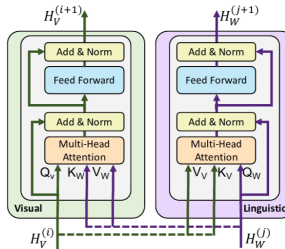
- Visual embeddings are passed to the Transformer with original text embeddings; the model implicitly discovers useful alignments between both sets of inputs and builds a new joint representation



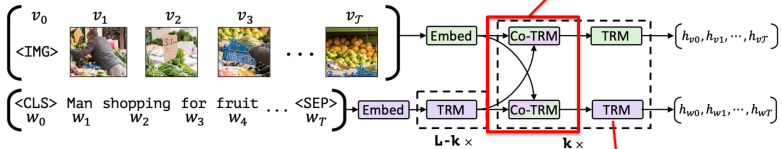
Source: [Li et al., 2019]

Slide source: [Li et al., 2019]

- Adds set of visual embeddings F , where each $f \in F$ corresponds to a bounding region in the image, derived from an object detector
- Embeddings are computed by summing:
 1. f_o , visual feature representation of bounding region of f , computed by a CNN
 2. f_s , a segment embedding to indicate it is an image embedding and not a text embedding
 3. f_p , a positional embedding to align words and bounding regions in the input



Cross-Modal Transformer Modules

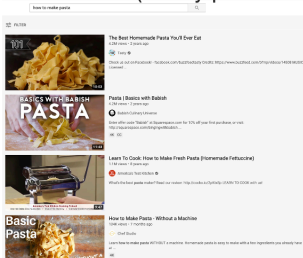


Unimodal Transformer

Source: [Lu et al., 2019]

- **Goal:** Learn better visual representations by taking advantage of large-scale video+language resources (co-learning)
- Instructional videos (weakly-paired data)

Instructional videos (weakly-paired data)



it's turning into a much thicker mixture



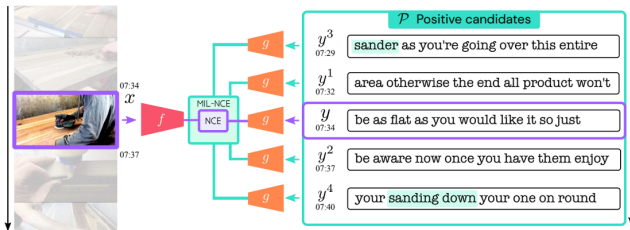
The biggest mistake is not kneading it enough



...

slide source: [Miech et al., 2019]

Input data: "a short 3.2 seconds video clip (32 frames at 10 FPS) together with a small number of words (not exceeding 16)"



How to handle this misalignment?

Multi-instance learning!

How to do it self-supervised?

Contrastive learning!

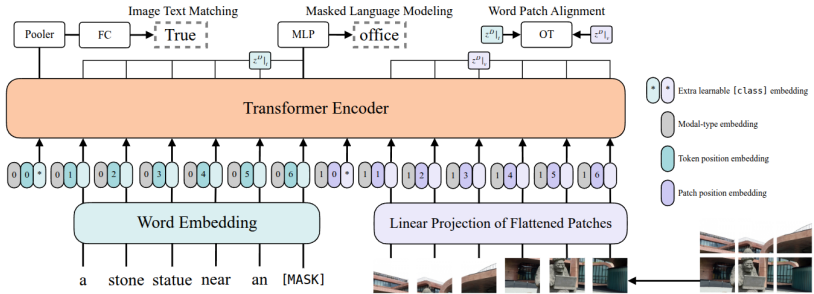
Objective: Given video x and text y from a positive set P_i and a negative set N_i , maximise the positive / total score ratio

$$\max_{f,g} \sum_{i=1}^n \log \left(\frac{\sum_{(x,y) \in \mathcal{P}_i} e^{f(x)^\top g(y)}}{\sum_{(x,y) \in \mathcal{P}_i} e^{f(x)^\top g(y)} + \sum_{(x',y') \sim \mathcal{N}_i} e^{f(x')^\top g(y')}} \right)$$

Note: Doing so requires maximizing $f(x)^\top g(y)$ for only positive examples

1. Using sets of positive and negative examples to "wash out" the misaligned text
2. Ideally, we would maximise all positives over all possible negatives (intractable)

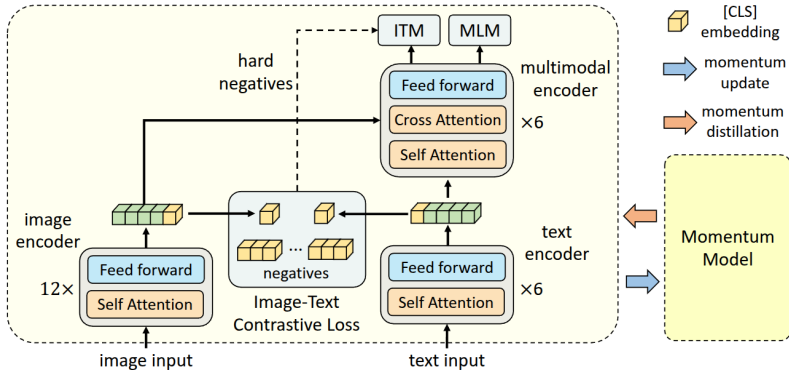
Vision-and-language transformer without convolution or region supervision [Kim et al., 2021]



Source: [Kim et al., 2021]

- Lightweight in both **language and vision encoder**, but heavy in **modality interaction**
- Masked Language Modelling (MLM) loss: Bert
- Word Patch Alignment loss is very time-consuming
- Read more for Image Text Matching (ITM) loss [Zhang and Lu, 2018]

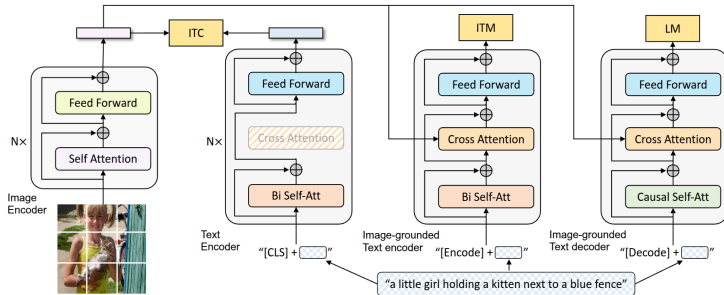
Align before Fuse: Vision and Language Representation Learning with Momentum Distillation [Li et al., 2021]



Source: [Li et al., 2021]

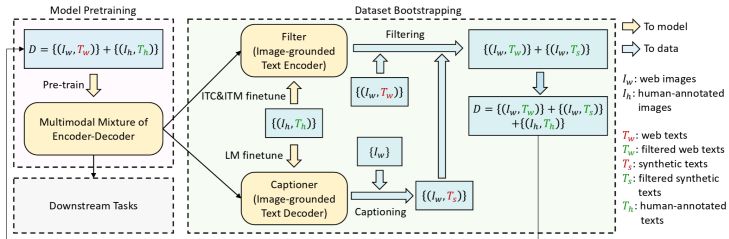
- Integration of MoCo [He et al., 2020] + ViT Vaswani et al. [2017] + Bert Devlin et al. [2019]
- The Image-Text Contrastive (ITC) Loss ensures the image and text embeddings align together before fed into the multimodal fusion module
- Use the **hardest negative image-text pair** (the negative sample that has the highest ITC loss) for ITM loss

Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation Li et al. [2022b]



Source: [Li et al., 2022b]

- Improved version of ALBEF [Li et al., 2021] with more language attention modules that **sharing weights**
- Causal Self-Attention module in the decoder is **single-directional** for generative tasks
- Filter-Captioner model for improving data quality



Source: [Li et al., 2022b]

Multimodal Learning

Multimodal Reasoning

Question

Is the skateboard airborne?

Image



Answer
yes

Source: [Lu et al., 2016]

How can we use attention?

Question

Is the skateboard airborne?

Image



Language can
be used to
attend the image

Answer
yes

Source: [Lu et al., 2016]

Question

Is the skateboard airborne?

Image



Image could
also be used to
attend the text

Answer
yes

Source: [Lu et al., 2016]

Question

Is the skateboard airborne?

Image



Or do both!

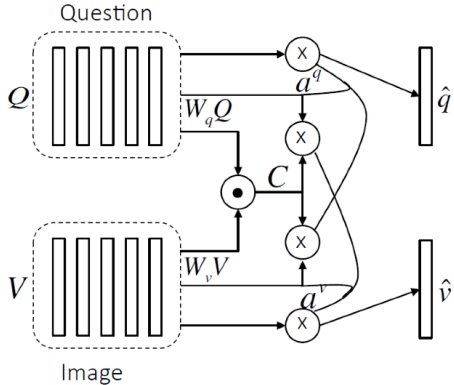
Answer

yes

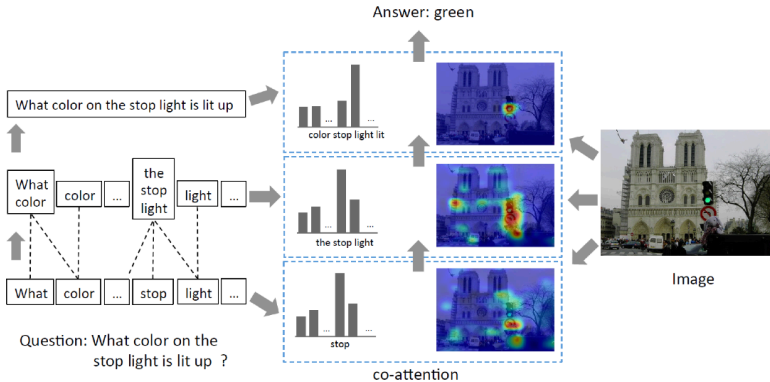
Source: [Lu et al., 2016]

Question

Is the skateboard airborne?

Image

Source: [Lu et al., 2016]

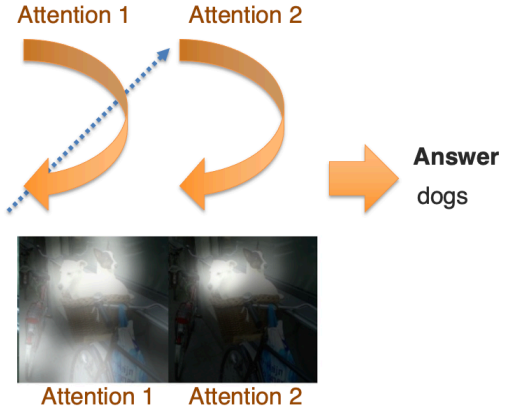


Source: [Lu et al., 2016]

Question

What are sitting in the basket on a bicycle?

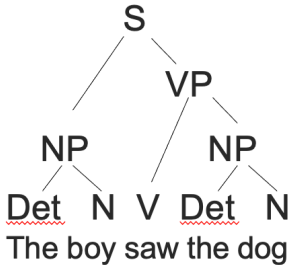
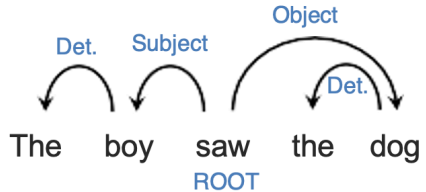
Image



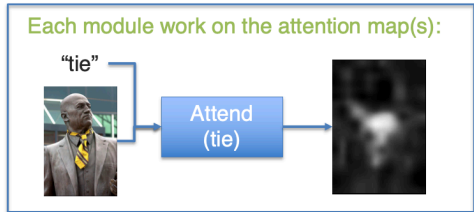
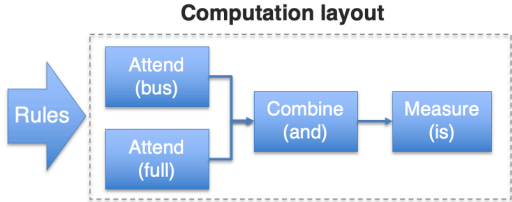
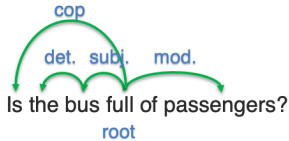
Source: [Yang et al., 2016]

- Bottom-up and top-down attention for image captioning and visual question answering [Anderson et al., 2018]
 - Adds the idea of object-based representations
- Bilinear Attention Pooling [Kim et al., 2018]
 - Extend low-rank bilinear pooling to multimodal
- Beyond bilinear: Generalized multimodal factorized high-order pooling for visual question answering [Yu et al., 2018]

**How can we make this attention process more interpretable?
Can we leverage prior knowledge (e.g., language structure)?**

**Constituency Parsing****Dependency Parsing**

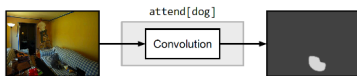
Source: L-P. Morency, CMU



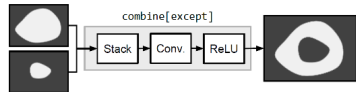
Source: [Andreas et al., 2015]

Analyse the image

$\text{attend} : \text{Image} \rightarrow \text{Attention}$

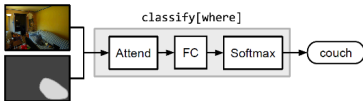


$\text{combine} : \text{Attention} \times \text{Attention} \rightarrow \text{Attention}$

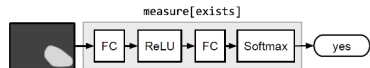


Make a prediction

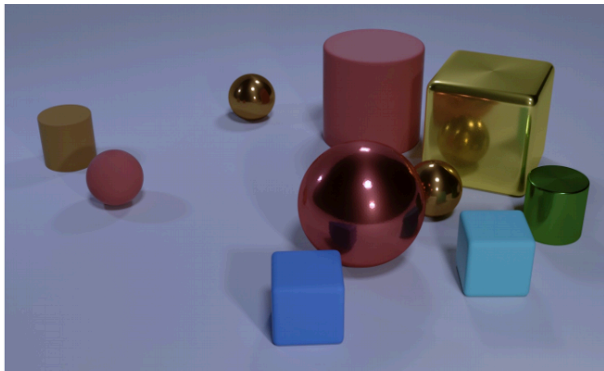
$\text{classify}[\text{where}]$



$\text{measure} : \text{Attention} \rightarrow \text{Label}$

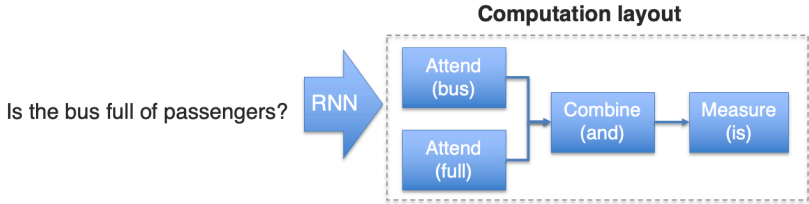


Source: [Andreas et al., 2015]



- Q: Are there an equal number of large things and metal spheres?
- Q: What size is the cylinder that is left of the brown metal thing that is left of the big sphere?
- Q: There is a sphere with the same size as the metal cube; is it made of the same material as the small red sphere?
- Q: How many objects are either small cylinders or metal things?

Source: [Johnson et al., 2017]



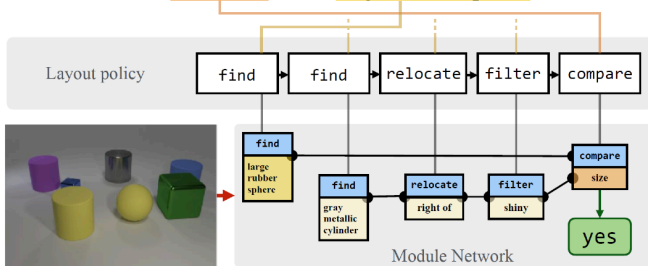
Source: [Hu et al., 2017]



Source: [Hu et al., 2017]

No need to parse the question!
No rule-based creation of the layout!

There is a shiny object that is right of the gray metallic cylinder;
does it have the same size as the large rubber sphere?



Source: [Hu et al., 2017]

Interactive Machine Learning

- Semi-supervised Learning
- Active Learning
- Other Interesting AL Techniques Used in Practice
- Graph-based Active and Semi-supervised Methods
- Deep Active Learning

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